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A framework for quantifying and leveraging uncertainty in pre-trained CT denoising model

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Abstract

Objective: To develop an architecture-agnostic framework that estimates, calibrates, and leverages total uncertainty (aleatoric + epistemic) in pre-trained, deep-learning denoising models for low-dose computed tomography (CT).

Methods: Aleatoric and epistemic uncertainties were estimated using physics-based inference-time augmentation and training-free, post-hoc Monte Carlo dropout, respectively, followed by non-parametric re-calibration for improved uncertainty calibration. To leverage uncertainty, we explored adaptive local fusion (ALF) guided by local mean-to-uncertainty ratio. For proof-of-concept, this framework was assessed using pre-trained U-net and ResNet-based models across datasets varying in CT tasks, radiation dose, and lesion characteristics. Uncertainty estimation and calibration were assessed in cadaver scans, using normalized-root-mean-square-error (NRMSE) and normalized-calibration-error (NCE), respectively. ALF was evaluated with chest and liver exams, using noise, structural similarity index (SSIM), and lesion detectability. Lesion detectability was quantified using clinically validated deep-learning model observer, with Wilcoxon signed-rank test to assess significance.

Results: This framework provided accurate uncertainty quantification and calibration: NRMSE range [1.2%, 2.4%], NCE [0.9%, 2.2%]. Compared to original pre-trained models, ALF yielded comparable or lower noise, improved lesion structural fidelity and detectability ($p<0.05$): For lung

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Code availability
We are open to sharing source code with researchers for non-commercial use, subject to our institutional approval.

nodules – noise reduction up to 69.7%, SSIM range (ALF vs pre-trained) [0.92, 0.96] vs [0.78, 0.87], detectability improvement up to 12.9%; for liver metastases – noise reduction up to 35.0%, SSIM range (ALF vs pre-trained) [0.82, 0.99] vs [0.80, 0.98], detectability improvement up to 13.2%.

Conclusion: Our framework effectively benchmarked and utilized total uncertainty to enhance diagnostic image quality with pre-trained CT denoising models.

Significance: This framework can facilitate performance monitoring, deployment optimization, and trustworthiness establishment.

Index Terms—

Computed tomography; deep learning; denoising; lesion detectability; uncertainty

I. INTRODUCTION

Computed tomography (CT) has revolutionized medicine by enabling high-quality imaging for diagnosis and treatment planning [1, 2]. The increased use of CT has raised concerns on excessive radiation exposure, prompting advances in CT denoising and reconstruction algorithms. Compared to iterative reconstruction (IR) algorithms, emerging deep-learning (DL) denoising algorithms show promise for further improving diagnostic image quality in computed tomography (CT). However, these algorithms may distort radio-pathological features or introduce new artifacts, particularly at aggressively reduced dose [3-5]. Like other DL techniques, these algorithms are also affected by aleatoric uncertainty (from noise / artifacts and intrinsic randomness in inputs) and epistemic uncertainty (from randomness in model weights) [6-9], which hinders reliable assessment of challenging lesions, limits radiation dose reduction, and undermines trust. Despite these concerns, information regarding the uncertainty of these denoising algorithms and actionable guidance are frequently absent in clinical settings. Therefore, critical needs exist for quantifying and leveraging uncertainty to facilitate continual performance monitoring, optimize algorithm deployment, and establish trustworthiness across clinical tasks.

Many prior studies in machine learning have focused on quantifying epistemic uncertainty, with more directed toward classification tasks than regression tasks. Examples are summarized as follows. Bayesian DL provides a generic framework that estimates epistemic uncertainty through sampling from posterior distribution, oftentimes assuming known parametric forms (e.g., Gaussian) of true posterior [10-12]. Gradient-based techniques provide scores to represent epistemic uncertainty using gradient information, i.e., the change of model outputs relative to that of model weights, which is frequently derived from deeper layers of DL models [13-16]. Monte Carlo (MC) dropout leverages dropout operation as a Bayesian approximation for quantifying epistemic uncertainty [17]. Deep ensembles empirically capture epistemic uncertainty by aggregating predictions across separately trained models [18]. Meanwhile, approaches to estimating aleatoric uncertainty include special-purpose methods developed specifically for this task, and other methods that were extended from the original task of epistemic uncertainty estimation. Several examples are presented as follows. Entropy-based methods capture aleatoric uncertainty via the entropy of softmax probability distribution [16, 19]. Quantile regression estimate conditional quantiles to reflect aleatoric uncertainty in model outputs [20]. Heteroscedastic regression parameterize the distribution (frequently as Gaussian) of model outputs with input-dependent variance [21]. Notably, these prior methods exhibit competitive performance, whereas various key challenges remain in practice. For instance, Bayesian DL cannot be readily applied to non-Bayesian models. The methods with parametric forms can underperform due to misalignment with actual distribution. Gradient-based and quantile-regression methods can sometimes conflate aleatoric and epistemic uncertainty [16, 20]. Other methods, such as deep ensembles and standard MC dropout, require careful re-training with original training datasets. Consequently, such methods cannot be readily applied to pre-trained models. Considering these caveats, there is no “one size fits all” uncertainty quantification method; rather, methods should be carefully selected and configured for each new scenario.

Several applications of uncertainty quantification have been investigated in general machine learning contexts, including uncertainty-aware decision making [22], active learning [23], selective prediction [24], and out-of-distribution detection [25], with potential for extension to medical imaging domain. In x-ray CT imaging, emerging studies have explored uncertainty quantification and / or utilization in denoising, reconstruction, or artifact correction tasks, many of which leverage well-established uncertainty quantification methods, e.g., Bayesian framework and standard MC dropout. For instance, Bayesian uncertainty quantification were integrated into deep unrolling-based denoising [26], applied to source-to-target domain alignment for unsupervised denoising [27], and used to guide bias-uncertainty trade-off in material decomposition [28]. Another study combined Bayesian framework with GAN and used epistemic uncertainty to guide the fusion between GAN outputs and filtered-back-projection for cone-beam CT artifact correction [17]. Moreover, diffusion model-based techniques can directly quantify intrinsic uncertainty through sampling the variance of model outputs [29-32]. Furthermore, one recent study integrated knowledge distillation with a Bayesian framework to estimate patient-specific uncertainty of non-transparent pre-trained denoising models (i.e., unknown architecture and training data) [8]. To date, many prior methods have been tightly coupled with specific DL model architecture and/or have required dedicated training procedures, limiting their applicability to pre-trained denoising models commonly used in clinical settings. Additionally, multiple prior approaches have not provided built-in components or have employed extra mechanisms to ensure the calibration quality of the estimated uncertainty. Notice that uncalibrated uncertainty quantification would raise multiple issues in downstream tasks, such as misleading confidence [33], undermined trust and interpretability [34], and poor Bayesian optimization [35]. In parallel, strategies for effectively leveraging uncertainty in CT denoising remain relatively underexplored.

In this study, we aimed to provide an architecture-agnostic framework to quantify uncertainty in pre-trained CT denoising models, leveraging uncertainty for further enhancing diagnostic image quality. Our approach differs from prior methods in the following aspects. First, aleatoric uncertainty was estimated through physics-based inference-time augmentation, guided by a generic CT image acquisition model, without assuming a pre-defined distribution in denoising model outputs. Second, epistemic uncertainty was quantified through specially-configured post-hoc MC dropout for pre-trained models. The similarity between dropout-imposed surrogate and true posterior was maximized through minimizing negative log likelihood, which involved dropout rate and dropout-dependent total uncertainty. Third, a non-parametric recalibration method was developed to improve uncertainty calibration quality, which adaptively rescaled the estimated epistemic uncertainty to align the estimated total uncertainty with the expected error. Finally, an adaptive local fusion (ALF) process, guided by local predictive mean and uncertainty ratio (MUR), was developed to further improve diagnostic image quality of the denoised images. In particular, the impacts on lesion detectability were objectively assessed in clinically relevant lesion localization tasks, using clinically validated deep-learning model observer (DLMO). Further details of our approach and experimental validation are presented in the following sections.

II. Methods

The proposed framework is illustrated in Fig. 1. Details of each component are presented in the following sections.

A. Quantification of aleatoric uncertainty

The aleatoric uncertainty in denoised CT images is mainly attributed to various realizations of CT noise and artifacts as well as intrinsic randomness in input CT images (e.g., locations of given anatomical features). It can be directly estimated via inference-time augmentation that emulates different realizations of inputs by applying perturbation to the testing input images. Of note, standard geometric and pixel value transformations that are often used for training data augmentation (e.g., cropping, elastic transformation, and histogram equalization) may not be feasible for inference-time augmentation, since these transformations are weakly related or irrelevant to CT data conditions in denoising tasks. Therefore, we developed physics-based inference-time augmentation, following a generic image acquisition model:

$$Y_d = \phi_{\theta^*}(X_n) = \phi_{\theta^*}(\mathcal{T}_\alpha(X_0 + \epsilon)) \tag{1}$$

$$Y = \mathcal{T}_\alpha^{-1}(Y_d) \tag{2}$$

where Y_d denotes the denoised images from the pre-trained DL denoising model ϕ_{θ^*} (with fixed weights θ^*), X_n is the spatially transformed, noise-corrupted CT images of ground truth X_0 with given locations and image orientations, ϵ is image noise, $\mathcal{T}_\alpha$ is the distribution of spatial transformations, and $\mathcal{T}_\alpha^{-1}$ denotes the inverse transformation. In this work, $\mathcal{T}_\alpha$ included Bernoulli distribution of random flipping along horizontal axis $\beta \sim Bern(0.5)$ and uniform distribution of in-plane translation $\Delta \sim U(-15, 15)$ pixels, to mimic random configurations of CT scan directions and patient positions. Furthermore, the characteristics of CT noise ϵ is dependent on patient anatomy, scanner-specific CT data acquisition techniques, CT scan protocol, and image reconstruction algorithms. To simulate the k^{th} noise realization ϵ_k, a simple physics-based calibration process was developed:

$$\epsilon_{0,k} = R_{recon}\left(P_\lambda\left(x, f_{prj}(Y_d)\right) - f_{prj}(Y_d)\right) \tag{3}$$

$$\epsilon_k = F^{-1}\left(F(\epsilon_{0,k}) \cdot f_m\left(\frac{|F(X_n - Y_d)|}{|F(\epsilon_{0,k})|}\right)\right) \tag{4}$$

where $P_\lambda(\cdot)$ denotes Poisson process with random variable x and the expectation $f_{prj}(Y_d)$, $f_{prj}(\cdot)$ is forward projection, $R_{recon}(\cdot)$ is filtered-back-projection with Ram-Lak filter, $\epsilon_{0,k}$ is un-calibrated initial guess of the calibrated k^{th} noise realization ϵ_k, $f_m(\cdot)$ denotes a 2D median filter that creates smoothly changed scaling factors to adjust the simulated noise power distribution in Fourier domain, and $F(\cdot)$ and $F^{-1}(\cdot)$ denote Fourier transform and its inverse. In clinical settings, the pre-trained CT denoising models undergo extensive validation before deployment. Thus, we treat Y_d as a practical surrogate of ground truth X_0 for generating noise realizations. For k^{th} noise realization, the noisy input is represented as $Y_d + \epsilon_k$ where Y_d approximated X_0. Notably, the physics-based noise calibration process mitigates the mismatch between Y_d and X_0, because it effectively aligns the simulated noise ϵ_k with the difference image $X_n - Y_d$ which contains the discrepancy $X_0 - Y_d$ and the true noise ϵ. f_m serves as a practical regularizer, to smoothly transform the "noise power envelope" of the simulated noise toward that of reference noise without forcing identical pixel structure in spatial domain. Finally, aleatoric uncertainty was represented as variance δ_a^2. A further discussion about inference-time augmentation is presented in Sec. IV.

B. Quantification of epistemic and total uncertainty

Standard Monte-Carlo dropout method [17] requires that the training of ϕ_θ includes dropout operators and l_2 regularization on θ. These prerequisites may not be met in practice. To tackle this challenge, we carried out training-free, post-hoc Monte-Carlo dropout, which is summarized as follows. The probabilistic distribution $p(y_d \mid x_n)$ of denoised images can be formulated as

$$p(y_d \mid x_n) = \int_\theta p_{\phi_\theta}(y_d \mid x_n, \theta)\, p(\theta \mid X_T, Y_T) \tag{5}$$

where $p(\theta \mid X_T, Y_T)$ is the intractable posterior distribution, and $\{X_T, Y_T\}$ denotes the training set. Monte-Carlo dropout can be used to approximate $p(\theta \mid X_T, Y_T)$ by effectively applying Bernoulli distribution $q(\theta \mid b)$ over model weights in inference [17]:

$$\hat{p}(y_d \mid x_n, b) = \int_\theta p_{\phi_\theta}(y_d \mid x_n, \theta) q(\theta \mid b) \tag{6}$$

where b denotes the set of dropout rates. To implement (6), the dropout operators were inserted back into the pre-trained model ϕ_θ and activated during inference. To approximate $p(\theta \mid X_T, Y_T)$ with $q(\theta \mid b)$, one could derive the optimal b through minimizing the Kullback-Leibler divergence between $p(y_d \mid x_n)$ and $\hat{p}(y_d \mid x_n, b)$. This is equivalent to minimizing the negative conditional log likelihood of $\hat{p}(y_d \mid x_n, b)$, since $p(y_d \mid x_n)$ does not depends on b:

$$\begin{aligned} b = \underset{\mathrm{b}}{\arg\min}\, KL(p(y_d \mid x_n) \parallel \hat{p}(y_d \mid x_n, b)) \equiv \\ \underset{\mathrm{b}}{\arg\min} - E_p[\log \hat{p}(y_d \mid x_n, b)] \end{aligned} \tag{7}$$

Using normality assumption and law of large numbers, (7) can be simply re-formulated as:

$$b = \underset{\mathrm{b}}{\arg\min} \sum \left(\frac{1}{2} \log \delta_{tot,b}^2 + \frac{1}{2\delta_{tot,b}^2} (y_d - \hat{y}_{d,b})^2 \right) \tag{8}$$

$$\delta_{tot,b}^2 = \delta_a^2 + \delta_{e,b}^2 \tag{9}$$

where δ_e^2, $\delta_{tot,b}^2$, and $\hat{y}_{d,b}$ denote the epistemic and total uncertainty, and the predictive mean measured via Monte-Carlo dropout in inference, respectively, δ_a^2 is the aleatoric uncertainty. Further derivation from (7) to (8) is presented in Appendix. The total uncertainty and predictive mean were estimated through marginalization across all sampled realizations of inputs X_n and model weights θ. The optimization problem in (8) can be numerically solved through grid searching across $b \in [0, 1]$. Of note, our approach does not require access to the original training dataset. Instead, the optimal b could be found using newly collected samples from target CT tasks. In this preliminary work, the same b was used in all dropout operators for simplicity. Moreover, a known form of posterior was imposed in (8), but it can deviate from ground truth across varying applications. Therefore, a re-calibration method was developed to further improve estimation accuracy (See Sec. II.C). Additionally, a further discussion about post-hoc MC dropout is presented in Sec. IV.

C. Uncertainty re-calibration method

The estimated total uncertainty δ_{tot} would be deemed as well-calibrated, when it is well aligned with the observed performance of the denoising model. In regression task (e.g., CT denoising), the ideally calibrated total uncertainty is expected to reflect the expected error [36], i.e., $\delta^2_{tot,b} = E\left[\|\hat{y}_d - y_{gt}\|^2\right]$. Due to the discrepancy between $p(\theta \mid X_T, Y_T)$ with $q(\theta \mid b)$, the initially estimated $\delta_{e,b}$ (and thus $\delta_{tot,b}$) can be mis-aligned. Therefore, we developed an uncertainty re-calibration method to further improve the alignment through rescaling $\delta_{e,b}$. This method is summarized as follows. The rescaling was formulated via establishing a functional mapping from $\delta^2_{tot,b}$ to $E\left[\|\hat{y}_d - y_{gt}\|^2\right]$. In this work, the mapping was created via non-parametric histogram binning, using a re-calibration dataset with paired samples $\left\{E\left[\|\hat{y}_d - y_{gt}\|^2\right], \delta^2_{tot,b}\right\}$. Notably, this re-calibration was conducted using validation datasets dedicated to target CT tasks. Given the joint equal-frequency histogram, bin-wise rescaling factors can be simply derived as:

$$1 = \frac{\sum E\left[\|\hat{y}_d - y_{gt}\|^2\right]}{\sum \delta^2_{a,j} + s_l \delta^2_{e,b,j}} \Rightarrow s_l = \frac{\sum E\left[\|\hat{y}_d - y_{gt}\|^2\right] - \sum \delta^2_{a,j}}{\sum \delta^2_{e,b,j}} \tag{10}$$

where s_l denotes the mean scaling factor at the l^{th} bin, and $y_{gt,j}$ is the label of the j^{th} sample. In this work, we used 100 bins. After re-calibration, the same binning and bin-wise rescaling factors were applied to testing samples. For each testing sample, the same binning scheme was used to construct the corresponding joint histogram. Within each bin, the associated scaling factor was used to adjust the uncalibrated epistemic uncertainty associated with the same bin. The calibrated total uncertainty was obtained by summing aleatoric uncertainty and the re-scaled epistemic uncertainty. These parameters can be applied to new scanners, protocols, or tasks, if CT data characteristics remain similar (e.g., CT number calibration and noise texture). Otherwise, new validation data should be acquired to ensure calibration quality.

D. Adaptive local fusion

We used mean-uncertainty-ratio (MUR) $\varphi_{\mathrm{mur}} = \frac{\hat{y}_d}{\delta_{tot}}$ as the primary test statistic to gauge the reliability of local outputs of denoising model ϕ_θ. Lower φ_{mur} indicates higher dispersion in ϕ_θ output distribution, suggesting worse reliability of the corresponding denoised images. In CT denoising, δ_{tot} is heteroscedastic and spatially variant. Therefore, we calculated φ_{mur} for small regions of interest (ROIs) sliding across the field-of-view. An ALF process was developed to process local regions via adaptively fusing the original noisy inputs with the local predictive mean:

$$y_{i,alf} = \omega x_{i,n} + (1 - \omega)\hat{y}_{i,d} \tag{11}$$

where $y_{i,alf}$ denotes the ALF output in the i^{th} ROI, $x_{i,n}$ and $\hat{y}_{i,d}$ denote the original input and the predictive mean, respectively, and ω is the spatially varying weighting factor driven by φ_{mur}. Note that the original local inputs were frequently reconstructed from standard physics-based analytic or iterative reconstruction. In this preliminary study, we developed three custom weighting functions for deriving ω:

$$\omega_s = (1 + \exp(-\varphi_{\text{mur}}^{-1}))^{-1} \tag{12}$$

$$\omega_t = 1 - \frac{\exp(2(\varphi_{\text{mur}} + 2)^{-1}) - 1}{\exp(2(\varphi_{\text{mur}} + 2)^{-1}) + 1} \tag{13}$$

$$\omega_p = \min((\varphi_{\text{mur}} + 1)^{-0.1}, 1) \tag{14}$$

where ω_s, ω_t, and ω_p denote different ω from the modified sigmoid, tanh, and clipped power functions, respectively.

E. Model observer

Lesion detectability was quantified using clinically validated deep-learning model observer (DLMO) [37-39]. The DLMO was directly operated on patient CT images to estimate the average performance of radiologists in target diagnostic tasks. The framework of DLMO included a pre-trained network ϕ_R (e.g., ResNet pre-trained using ImageNet), a partial-least-square discriminant analysis (PLS-DA), and an internal noise component. The latent features from ResNet were engineered to mimic the response of human primary visual cortex, using the test statistics from PLS-DA:

$$\lambda_{0,x_i} = \Gamma_{\phi_R(x_i)} \cdot \beta_{PLS} \tag{15}$$

where λ_{0,x_i} is the initial test statistics mimicking human visual response at local region x_i, $\Gamma_{\phi_R(x_i)}$ denotes the latent features retrieved from ϕ_R and β_{PLS} indicates the template from PLS-DA. A lesion search process (e.g., sliding window of ROIs) is deployed to generate pixel-wise test statistics. The internal noise component was used to mimic human performance variability in lesion detection:

$$\lambda_{x_i} = \lambda_{0,x_i} + \gamma \cdot n \tag{16}$$

where λ_{x_i} is the final test statistics at x_i, $n \sim N\left(0, \sigma_{\lambda_{0,bkg}}\right)$ is a Gaussian random variable with zero mean and standard deviation equal to that of λ_0 in lesion-free background, and γ is a calibration factor determined via matching the levels of DLMO and human performance. The performance was gauged using the area under localization receiver operating characteristic curve A_L. The calibration of γ was carried out in one pre-selected experimental condition, and then the same value of γ was used in other conditions that involved varying configurations of radiation dose, image types and / or lesion characteristics (e.g., different diseases, size, contrast). To measure lesion detectability in ALF-derived images, the lesion search process was synchronized with the ROI sliding in ALF.

F. Experimental evaluation methods

For proof of concept, the proposed architecture-agnostic framework was evaluated using recent U-net and ResNet-based denoising models [40-42]. The U-net model was pre-trained for whole-body low-dose skeletal survey. The ResNet model was pre-trained separately for patient chest and abdominal CT exams. No dropout was used in pre-training. Model weights were optimized through an objective function that comprised feature reconstruction loss [43-45] and mean-squared-error. During inference, dropout operators were inserted into the pre-trained model. The optimal dropout rate and bin-wise rescaling factors were derived using validation datasets and thereafter applied to independent testing datasets. Further details of neural network models, CT datasets, and pre-training process are presented in the Appendix.

A cadaver dataset was collected to validate the quality of uncertainty quantification and calibration, using the same scan and reconstruction protocols from [40]. Due to lack of lesion annotation, lesion detectability was not assessed using this dataset. Cadaver specimens were repeatedly scanned 100 times to acquire varying realizations of noisy inputs, using the same scan protocol of whole-body low-dose skeletal survey. See the appendix for further details of this dataset. Model outputs were collected across repeated scans and four anatomical regions (denoted as “shoulder”, “lung”, “liver”, and “kidney” for convenience). The average cadaver images served as the surrogates of ground truth. The quality of inference-time augmentation was assessed using structural similarity index (SSIM) and normalized root-mean-square-error (NRMSE):

$$NRMSE = \frac{1}{M} \frac{\sum_m \left(\delta_{a,m,1}^2 - \delta_{s,m}^2\right)^2}{\max \delta_s^2 - \min \delta_s^2} \tag{17}$$

where $\delta_{a,m,1}^2$ is the predictive variance in m^{th} pixel from single CT scan, which is estimated using the proposed physics-based inference-time augmentation (Sec. II. A). $\delta_{s,m}^2$ is the measured variance from 100 repeated scans, and M denotes the total number of pixels. The quality of total predictive uncertainty δ_{tot}^2 was assessed using calibration plots (δ_{tot}^2 vs. $E\left[\left\|\hat{y}_d - y_{gt}\right\|^2\right]$) and normalized calibration error (NCE):

$$NCE = \frac{1}{L}\sum_{l} \frac{\mid \sum_{j} E\left[\|\hat{y}_{d} - y_{gt}\|^{2}\right] - \sum_{j} \delta_{tot,\,j}^{2} \mid}{\max E\left[\|\hat{y}_{d} - y_{gt}\|^{2}\right] - \min E\left[\|\hat{y}_{d} - y_{gt}\|^{2}\right]} \tag{18}$$

where L denotes number of bins, and y_{gt} denotes the ground truth. Notice that NCE is similar to the expected calibration error used in the other classification and regression tasks [36].

Patient chest and abdominal cases were used to assess the performance of ALF. Testing patient cases were retrieved from prior lung nodule [38] and liver metastases [39] localization tasks. Retrospective use of patient images was approved by Institutional Review Board (IRB # 12-007246). The lung nodule dataset included 10 experimental conditions, involving 4 radiation dose levels (10%, 25%, 50%, and 100% routine dose), 3 nodule size (3.4 mm, 5.4 mm, and 7.4 mm diameter), and 2 nodule types (part-solid and ground glass). The liver metastases dataset included 7 conditions, involving 3 dose levels (25%, 50%, and 100% routine dose), 3 lesion size (7 mm, 9 mm, and 11 mm), and 3 lesion contrasts (15 HU, 20 HU, and 25 HU). In both datasets, low dose exams previously were generated through inserting calibrated Poisson noise into real projection data from CT system, using clinically validated projection-domain noise insertion [46]. Also, lesion-present cases were previously generated through inserting target lesions at random anatomical locations, using clinically validated projection-domain lesion insertion [47, 48]. In lesion-present cases per condition and task, the same target lesions (i.e., consistent size, contrast, and types) were used in insertion to facilitate task-driven assessment in well-controlled experimental conditions. Each condition included 100 lesion-present / -absent cases. Local image noise levels were compared at relatively uniform anatomical regions (e.g., muscle, fat, or liver parenchyma). For each patient case, noise was measured from square region-of-interests (15×15 pixels) placed in the same anatomical region across all image types. Boxplots were used to assess the distribution and the trends of lesion-wise MUR across experimental conditions. SSIM of mean denoised lesion images with and without ALF was compared across all conditions, to evaluate the level of systematic distortion in denoised lesion appearance. For each condition, SSIM was calculated using the average denoised lesion images across all lesion locations, with the corresponding mean un-denoised routine-dose lesion images as the reference. Since lesions were in relatively uniform lung and liver parenchyma [38, 39], averaging largely “cancel out” surrounding background, local noise and artifacts. Consequently, SSIM predominantly captured systematic bias, i.e., the deviations in the expected model outputs from the practical surrogate of ground truth. Visual information fidelity (VIF) was further evaluated over individual lesion images, using the same reference images in SSIM calculation. The difference of VIF scores was tested using Wilcoxon signed rank test ($p<0.05$ as statistical significance). Furthermore, lesion detectability with and without ALF was compared using task-specific DLMOs. These DLMOs were previously calibrated and validated with fellowship-trained radiologists in lung nodule and liver metastases localization tasks. In this study, the same $\Gamma_{\phi_R(x_i)}$, β_{PLS}, and γ from [38, 39] were used. The potential gain of lesion detectability was separately evaluated across the three weighting functions of ALF HYPERLINK (12) - (14). Lesion detectability corresponding to pre-trained models served as the baseline in evaluation. Performance difference was tested using Wilcoxon signed rank test (P-value <0.05 as statistically significant).

III. Results

A. Evaluation in cadaver dataset

The physics-based inference-time augmentation enabled high-quality estimation of aleatoric uncertainty. The simulated noisy images exhibited noise texture and frequency characteristics comparable to original noisy images (Supplementary Fig. 1). Across different anatomical regions, the predictive variance from single scan resembled the reference measured from repeated CT scans (Fig. 2). The corresponding range of NRMSE and mean SSIM per image were 1.2% – 2.4% and 0.90 – 0.99, respectively (TABLE I). SSIM maps confirmed strong agreement between the reference and predicted variance (Supplementary Fig. 2). Moreover, the proposed re-calibration method effectively improved the quality of calibration, mitigating the underestimation of total uncertainty (Fig. 3). Range of NCE was 3.6% – 7.5% and 0.9% – 2.2% before and after calibration, respectively (TABLE II).

B. Evaluation in lung nodule dataset

Example images from a nodule-present case are illustrated in Fig. 4. The finer anatomical structures (e.g., nodules and small vessels) yielded lower MUR, compared to prominent ones, indicating worse vulnerability to uncertainty. Compared to pre-trained ResNet model, ALF reserved higher frequency components in the denoised images and exhibited relatively fewer residual structures and bias in the difference images. Moreover, the original pre-trained ResNet could distort the appearance of target nodule, leading to low DLMO response at nodule location and subsequent misdetection (Supplementary Fig. 3). In contrast, ALF yielded stronger DLMO response at target nodule locations. Compared to original noisy inputs, ALF reduced image noise across all dose levels ($p<0.05$ for all three weighting functions; Table III): range of mean percent image noise reduction (across 10%, 25%, 50% and 100% dose) [50.0%, 69.7%], [42.1%, 62.5%], [37.6%, 64.9%], [35.0%, 64.0%], respectively.

The nodule-wise MUR decreased as radiation dose level decreased (Appendix Fig. 1): median MUR 0.7, 1.6, 2.8, and 3.9 from 10% to 100% dose. No apparent dependency on nodule size or types was observed. Also, the large variation of nodule-wise MUR at routine dose indicated the heterogeneous impacts of immediate anatomical background on the reliability of denoised features. Furthermore, the ALF-derived images exhibited higher SSIM scores, compared to the pre-trained ResNet (Table IV). This indicated that the pre-trained model induced systematic bias while ALF improved image fidelity. Across individual lesions, ALF provided statistically significant gains in VIF compared with the pre-trained model (Appendix Table I).

Compared to the pre-trained ResNet, ALF yielded comparable or further improved mean lesion detectability, particularly in challenging conditions that involved small nodules and / or lower radiation dose levels (Fig. 5). The range of mean A_L improvement was [1.2%, 8.6%], [0.7%, 10.8%], [0.8%, 12.9%] using the modified sigmoid, tanh, and clipped power functions respectively ($p<0.05$). Additionally, the denoising model with and without ALF consistently improved lesion detectability relative to original iterative reconstruction whose lesion detectability was reported in [38], with mean A_L gains ranging from 3% to 37% in challenging conditions.

C. Validation in liver metastases dataset

Example images from a lesion-present case at 50% dose are illustrated in Fig. 6. Like the chest CT cases, lower MUR was observed at finer anatomical structures (e.g., lesions and small vessels), showing higher local uncertainty. Compared to the pre-trained model, ALF showed comparable or stronger noise reduction in the denoised images. Also, ALF exhibited much fewer residual structures and bias in the difference images. Moreover, the ALF showed less false positives in DLMO heatmaps. Compared to original noisy inputs, ALF reduced image noise across three dose levels ($p<0.05$; Table V): range of mean percent noise reduction (25%, 50%, 100% dose) [4.6%, 31.4%], [5.0%, 35.0%], [3.6%, 35.0%], respectively.

Like the findings in chest CT cases, the lesion-wise MUR also decreased as radiation dose level decreased (Appendix Fig. 2): median MUR 1.2, 1.8, and 4.0 from 25% to 100% dose. Lesion size or contrast did not exhibit apparent impacts on lesion-wise MUR. Given large variation of lesion-wise MUR at routine dose level, local anatomical background also demonstrated heterogeneous impact on local reliability of denoised pathological features. Moreover, the mean lesion images from ALF and the pre-trained ResNet showed similar SSIM scores at routine dose, whereas ALF yielded higher SSIM at 50% and 25% dose levels (Table VI). This suggested that the pre-trained denoising model systematically distorted lesion images at lower dose while ALF mitigated this issue. Across individual lesions, ALF with m-Sigmoid and m-Tanh achieved statistically significant gains in VIF compared with the pre-trained model (Appendix Table II).

Compared to the pre-trained model, ALF exhibited similar or improved mean detectability of metastatic liver lesions, particularly in challenging conditions that involved smaller lesion size, lower contrast, and / or lower radiation dose (Fig. 7). The range of mean A_L improvement was [1.7%, 13.2%], [1.1%, 6.0%], [1.1%, 12.9%] using the modified sigmoid, tanh, and clipped power functions respectively ($p<0.05$). Additionally, the pre-trained model without ALF degraded lesion detectability compared with original iterative reconstruction whose detectability was reported in [39], with mean A_L loss of 3% to 22%. In contrast, incorporating ALF method reduced the loss of detectability at lower dose (up to 13%), while further improved lesion detectability of smaller, lower contrast lesions at routine dose, with mean A_L gains of 2% to 7% relative to original iterative reconstruction.

IV. Discussion

We have presented an architecture-agnostic framework to provide calibrated uncertainty estimation for pre-trained CT denoising models and to leverage uncertainty for enhancing diagnostic image quality. In preliminary validation, it was evaluated using two pre-trained denoising models across cadaver and patient scans. Our approach exhibited high-quality uncertainty estimation and calibration, reduced noise and systematic bias, and improved lesion detectability in clinically relevant tasks.

The proposed framework can benchmark uncertainty level for continual patient- and task-wise performance monitoring of clinically deployed CT denoising models. Excessive uncertainty could indicate unreliable image quality, suggesting the need for additional expert review on radiological findings or scan parameters. The experts' feedback can, in turn, help establish actionable guidance for model management. Additionally, ALF can further enhance diagnostic image quality of denoised images.

The proposed physics-based inference-time augmentation provided CT-domain-specific realizations of noisy inputs, thereby improving the relevance and trustworthiness of the estimated aleatoric uncertainty in clinical CT tasks. This contrasts with prior data-fitting methods (e.g., heteroscedastic regression) which need extra training and could be potentially overfitted. Moreover, our approach employed a physics-based calibration process to approximate scanner-specific noise realizations in forward-projected sinogram and image domains, without requesting real projection data of CT systems. This noise calibration process effectively reduces the approximation errors from using original denoised images Y_d as a surrogate of ground truth X_0, and the subsequent uncertainty re-calibration further suppresses the impacts of remining errors. Additionally, a recent physics-informed DL-based image domain noise insertion [49] may be employed to further enhance the realism of simulated CT noise texture. Additionally, projection-domain noise insertion could also be used when real projection data of CT scanner is accessible [7].

To date, two prior studies have explored post-hoc MC dropout for estimating epistemic uncertainty in pre-trained models originally trained without dropout. The true posterior is commonly intractable. Since the post-hoc MC dropout enabled a high-quality approximation of the true posterior (Sec. II. B.), the resulting epistemic uncertainty served as a practical surrogate for ground truth. The differences between our approach and the prior studies are summarized as follows. One study applied it to autonomous driving, while adopting an assumption that the output distribution of pre-trained model well approximated the ground truth in training samples [50]. Of note, this strong assumption could fail in diverse real-world conditions, since approximation accuracy depends on model capacity [51] and the quality of training process. Also, the corresponding approach requested access to original training datasets and lacked explicit mechanisms to ensure uncertainty calibration quality. Another recent study used a similar strategy for computer vision tasks such as crowd counting, while introduced an unknown constant scaling factor to improve calibration quality through an optimization process similar to Sec. II.B [52]. Notably, this prior approach still imposed Gaussian-form distributions in joint uncertainty estimation and re-calibration, which can be inadequate for approximating non-Gaussian true posterior. In contrast, we independently explored and configured post-hoc MC dropout for CT denoising tasks. Our approach differed from these prior works in two key aspects. First, our approach did not rely on ideal training outcome or original training datasets. Second, our approach provided an adaptive, non-parametric re-calibration method to ensure calibration quality, rather than solely using a specific posterior form. Practically, our approach could complement the prior methods in scenarios where the aforementioned conditions are not strictly satisfied.

Aleatoric and epistemic uncertainty are commonly treated as independent and additive components of total uncertainty, since source independence holds true in common applications [21]. This enables a practical way to separately assess both types of uncertainty. In CT denoising, the independence assumption could be violated when aleatoric and epistemic uncertainty intertwined. Given high noise at small structures, the uncertainty of target model's knowledge about whether those structures exist could be influenced by the noise. Nonetheless, the proposed method is expected to enable high quality estimation of total uncertainty in target tasks, once it is well-calibrated. The re-calibration process may be needed for newer tasks, if the corresponding image characteristics are drastically different from the original tasks.

Three variants of weighting functions (i.e., (12) - (14)) were explored within the proposed ALF. These functions were empirically configured, inspired by popular activation functions and data transformation, to introduce adaptive fusion across varying mean-uncertainty-ratio. Using these weighting functions, the ALF constrained image noise to fall between that of local predictive mean and original noisy images. The resulting noise level in ALF images varies relative to that of original denoising model without ALF. Analyses of lesion detectability revealed that the relative performance gains were heterogenous across experimental conditions. This reflects the combined effects of lesion characteristics and the trade-off between residual noise and bias imposed by specific weighting functions. Notably, either excessive residual bias or noise can constrain performance gain. For instance, in chest CT cases with PSN, the ALF with m-Sigmoid produced relatively more enhancement in lesion detectability at 25% dose than at 50% dose. At higher dose, the m-Sigmoid tends to incorporate a larger contribution of local predictive mean during blending, which could potentially introduce more residual bias, if any, from local predictive mean at lesion locations and consequently reduce performance gain. These findings highlighted the necessity for task-driven optimization in the theoretical design or selection of weighting functions. Also, this represents a promising direction in our future research. For a given pre-trained model and clinical task, an appropriate weighting function may be chosen based on its combined performance in noise reduction, systematic bias control, and overall gain in lesion detectability across experimental conditions. Accordingly, m-Power is preferred for the presented lung nodule searching task, while m-Sigmoid and m-Tanh are more suitable for the presented liver lesion searching task.

We acknowledge several limitations in this work. First, the proof-of-concept validation only involved two pre-trained denoising models. Nonetheless, our approach is architecture-agnostic and can be extended to other neural network models. Second, the proposed ALF was only assessed in lung nodule and liver metastases localization tasks. Of note, these tasks were selected as litmus tests, given inherent challenges involving small, low-contrast lesions and / or lower radiation dose. Extending the evaluation to additional clinical CT tasks would be an important direction for future work but beyond the scope of current study. Third, the experimental evaluation only involved validation datasets from single site. This was primally due to limited access to multi-site low-dose CT datasets. Finally, further validation with additional DL denoising models, pathological characteristics, radiation dose levels, CT scan protocols, CT systems, and larger patient cohort are warranted in our follow-up studies.

V. Conclusion

An architecture-agnostic framework was developed to explicitly quantify, calibrate, and leverage patient-specific aleatoric and epistemic uncertainty for pre-trained CT denoising models by integrating physics-based inference-time augmentation, post-hoc Monte Carlo dropout, non-parametric uncertainty calibration, and adaptive local fusion. As a proof-of-concept, it was validated using pre-trained U-net and ResNet-based models – originally trained without dropout – with repeated cadaver CT scans and clinically relevant lesion searching tasks. The proposed framework exhibited potential for benchmarking uncertainty, continual performance monitoring, optimizing algorithm deployment, and improving trust across varying clinical tasks.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

Acknowledgment

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Appendix

A. Derivation

The derivation from Equation (7) to (8) is further explained as follows. As illustrated in (7), the optimal dropout rate is derived through minimizing the negative expectation of conditional log probability. Using the law of large numbers, this expectation can be approximated with the sample mean:

$$b \equiv \underset{b}{\arg\min} - E_p[\log \hat{p}(y_d \mid x_n, b)] \approx \underset{b}{\arg\min} - \frac{1}{K} \sum \log \hat{p}(y_{d,k} \mid x_{n,k}, b) \tag{19}$$

where K denotes number of samples, $x_{n,k}$ and $y_{n,k}$ are the k^{th} sample of noisy input and the denoised images. Assuming normality, the equation (19) can be further simplified using normal form of $\hat{p} \sim \mathcal{N}(\hat{y}_{d,b}, \delta^2_{tot,b})$:

$$b = \arg\min_{\mathrm{b}} - \frac{1}{K} \sum \log\left(\frac{1}{\sqrt{2\pi\delta_{tot,b}^2}} \exp\left(-\frac{(y_{d,k} - \hat{y}_{d,b})^2}{2\delta_{tot,b}^2}\right)\right) = \arg\min_{\mathrm{b}} \frac{1}{K} \sum \frac{(y_{d,k} - \hat{y}_{d,b})^2}{2\delta_{tot,b}^2} + \frac{1}{2}\log\delta_{tot,b}^2 + \log 2\pi \tag{20}$$

The constant term $\log 2\pi$ drops out during optimization. Finally, K and k can be ignored for further simplification, which arrives at the formulation in (8).

B. Denoising models

Several common settings were used for U-net and ResNet models. Both were implemented using TensorFlow. The input layer used 3 channels which corresponded to 3 consecutive noisy images from adjacent slices. The output layer only had one channel which corresponded to the middle image. A bypass connection was added between the output and the input middle image to enable residual learning. All convolutional layers used 3×3 kernels. Stride was fixed at 1×1. Batch-normalization and Relu activation were applied after each convolutional layer. Other hyper-parameters followed the default settings. Additional settings are summarized as follows. In U-net model, each convolutional layer used 64 channels. The encoder side employed 3 max pooling layers with pool size 2×2, and the decoder side employed 3 up-sampling layers with nearest-neighbor interpolation. In ResNet model, six residual blocks were included, where each convolutional layer used 128 channels. No pooling was employed.

C. CT datasets

For U-net model, the training set was acquired from patient whole-body low-dose CT exams, including ~50,000 paired training samples. See [41] for details of training set. The cadaver dataset was collected using the same scan and reconstruction parameters used in patient exams: 120 kV, 70 quality reference mAs, CareDose4D, Siemens Flash system; filtered-back-projection with sharp reconstruction filter Br64, slice thickness / increment 3/3 mm. Repeated CT scans (n=100) were performed for cadaver specimens. Stratified splitting was used to re-group cadaver CT images from four anatomical regions (i.e., “Shoulder”, “Chest”, “Liver” and “Kidney”) into validation and testing sets. The validation set was used in uncertainty re-calibration, including >2,000 paired patches (64×64 pixels per patch) per scan. The independent testing set was used to assess the quality of uncertainty estimation and calibration, including 36 full field-of-view images (512×512 pixels per image) per scan.

For ResNet model, the training and validation sets were previously acquired from independent patient cases that were excluded from testing sets. For the lung nodule localization task, the training and validation sets were established from patient chest CT exams at routine dose: 120 kV, 50 quality reference mAs, Siemens Force; commercial iterative reconstruction (ADMIRE, Siemens) with medium filter Bv49. The training and validation sets included ~60,000 and ~5,000 paired image patches (64×64 pixels per patch),

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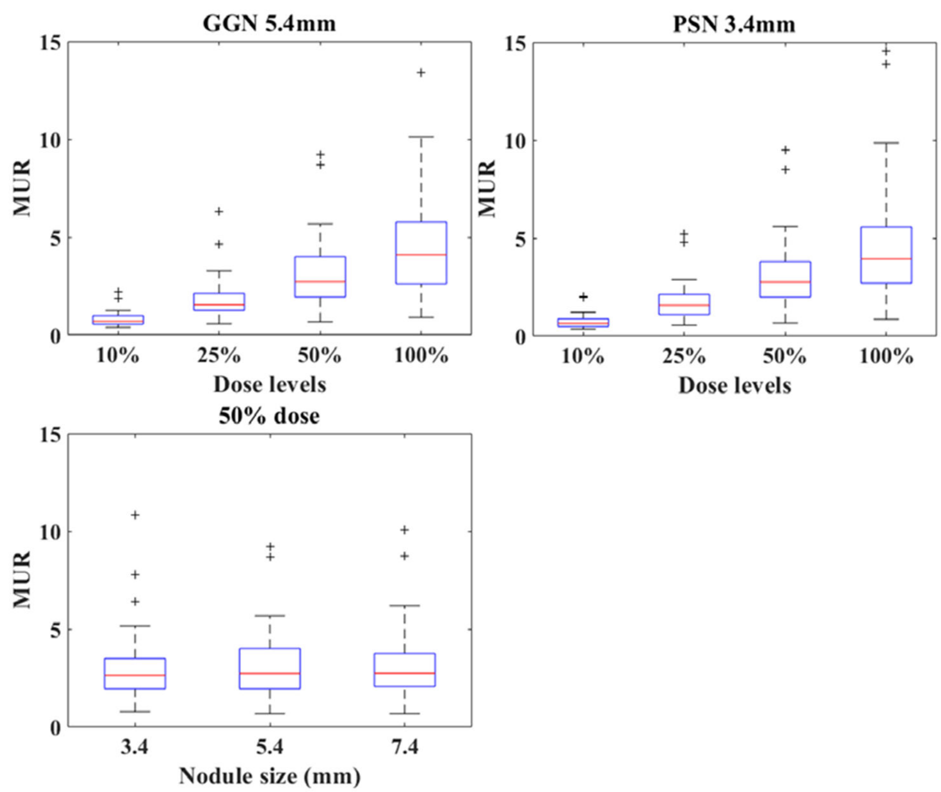
respectively. The independent testing sets comprised trial cases across 10 experimental conditions (Sec. II.F). Each trial case included the images of one lung per testing patient case (512×256 pixels per image). For the liver metastases localization task, the training and validation sets were similarly prepared. Original abdominal exams were conducted following clinical routine protocol: 100 / 120 kV, 240 quality reference mAs, CareDose4D, Siemens Flash system; commercial iterative reconstruction (SAFIRE, Siemens) with medium filter I30 and slice thickness / increment 3/3 mm. The independent testing sets comprised trial cases across 7 conditions (Sec. II.F). Each trial case included the images of one testing liver segment (160×160 pixels per image). In both tasks, the ResNet models were pre-trained using the paired 25% and routine dose images, while thereafter deployed to process images at all radiation dose levels.

D. Pre-training process

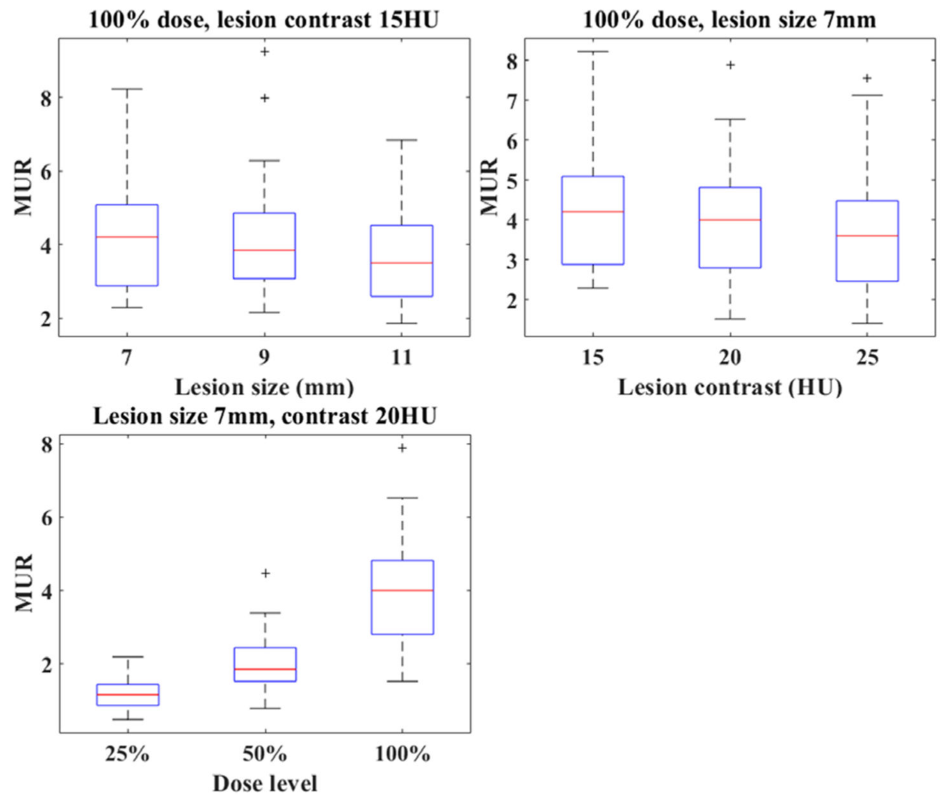
Both U-net and ResNet model were previously implemented in TensorFlow. No dropout was used in pre-training. Model weights were optimized using an objective function that comprised mean-squared-error and feature reconstruction loss. Adam optimizer was used with scheduled learning rate decay (0.001 to 0.00001). In pre-training, the validation samples were primarily used to monitor convergence and overfitting. No additional fine-tuning was performed.

E. Results



Appendix Fig. 1.

Boxplots of mean-uncertainty-ratio (MUR) per lung nodule across different experimental conditions.



Appendix Fig. 2.

Boxplots of mean-uncertainty-ratio (MUR) per liver lesion across different experimental conditions.

Appendix Table I

ASSESSMENT OF INDIVIDUAL LUNG NODULE IMAGE QUALITY USING VIF (MEAN ± STANDARD DEVIATION) ACROSS EXPERIMENTAL CONDITIONS

	5.4mm GGN 10%	5.4mm GGN 25%	5.4mm GGN 50%	5.4mm GGN 100%	3.4mm PSN 10%	3.4mm PSN 25%	3.4mm PSN 50%	3.4mm PSN 100%
Un-denoised	0.84±0.06	0.85±0.05	0.85±0.06	/	0.84±0.05	0.85±0.06	0.85±0.06	/
m-Sigmoid	0.84±0.05	0.84±0.05	0.83±0.06	0.84±0.06	0.83±0.06	0.84±0.05	0.83±0.06	0.83±0.06
m-Tanh	0.83±0.05	0.83±0.06	0.83±0.06	0.83±0.06	0.84±0.06	0.84±0.06	0.83±0.06	0.83±0.06
m-Power	0.84±0.05	0.84±0.06	0.84±0.06	0.83±0.06	0.83±0.06	0.83±0.06	0.83±0.06	0.83±0.06
No ALF[†]	0.82±0.05	0.82±0.05	0.81±0.06	0.81±0.06	0.80±0.05	0.80±0.06	0.80±0.06	0.80±0.07

Visual information fidelity (VIF) was derived using individual lesion image across all lesion locations per experimental condition. The un-denoised mean routine-dose lesion images per condition served as the reference. The 3.4mm and 7.4mm GGN were not involved in comparison due to lack of routine-dose references.

“Un-denoised” denotes the original reconstruction (without deep-learning denoising) across the four dose levels.

“m-Sigmoid”, “m-Tanh”, “m-Power” correspond to the ALF with weighting functions (12) - (14), respectively.

"No ALF" corresponds to the original pre-trained denoising model.

†In each condition, "No ALF" shows statistically significantly lower VIF than the other image types across all conditions (p<0.05), while no statistically significant difference was observed among the other types.

Appendix Table II

ASSESSMENT OF INDIVIDUAL LIVER LESION IMAGE QUALITY USING VIF (MEAN± STANDARD DEVIATION) ACROSS EXPERIMENTAL CONDITIONS

	7mm 15HU 100%	7mm 20HU 100%	7mm 20HU 50%	7mm 20HU 25%	7mm 25HU 100%	9mm 15HU 100%	11mm 15HU 100%
Un-denoised	／	／	0.67±0.05	0.58±0.07	／	／	／
m-Sigmoid*	0.77±0.05	0.80±0.04	0.73±0.04	0.66±0.07	0.79±0.06	0.80±0.04	0.80±0.06
m-Tanh*	0.77±0.05	0.79±0.04	0.73±0.04	0.66±0.07	0.79±0.06	0.81±0.04	0.80±0.06
m-Power	0.75±0.06	0.75±0.05	0.67±0.05	0.59±0.07	0.75±0.06	0.76±0.05	0.76±0.06
No ALF	0.74±0.06	0.75±0.06	0.68±0.06	0.59±0.07	0.76±0.07	0.77±0.05	0.77±0.07

Visual information fidelity (VIF) was derived using individual lesion image across all lesion locations per experimental condition. The un-denoised mean routine-dose lesion images served as the reference.

"Un-denoised" denotes the original reconstruction (without deep-learning denoising) across the four dose levels.

"m-Sigmoid", "m-Tanh", "m-Power" correspond to the ALF with weighting functions (12) - (14), respectively.

"No ALF" corresponds to the original pre-trained denoising model.

*"m-Sigmoid" and "m-Tanh" exhibit statistically significantly higher VIF than the other image types across all conditions (p<0.05)

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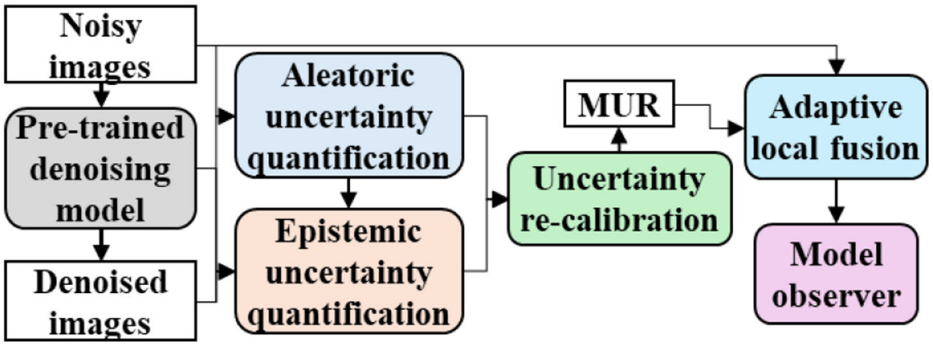

Fig. 1.

The proposed framework to quantify and leverage uncertainty in pre-trained denoising model. MUR – mean-uncertainty-ratio.

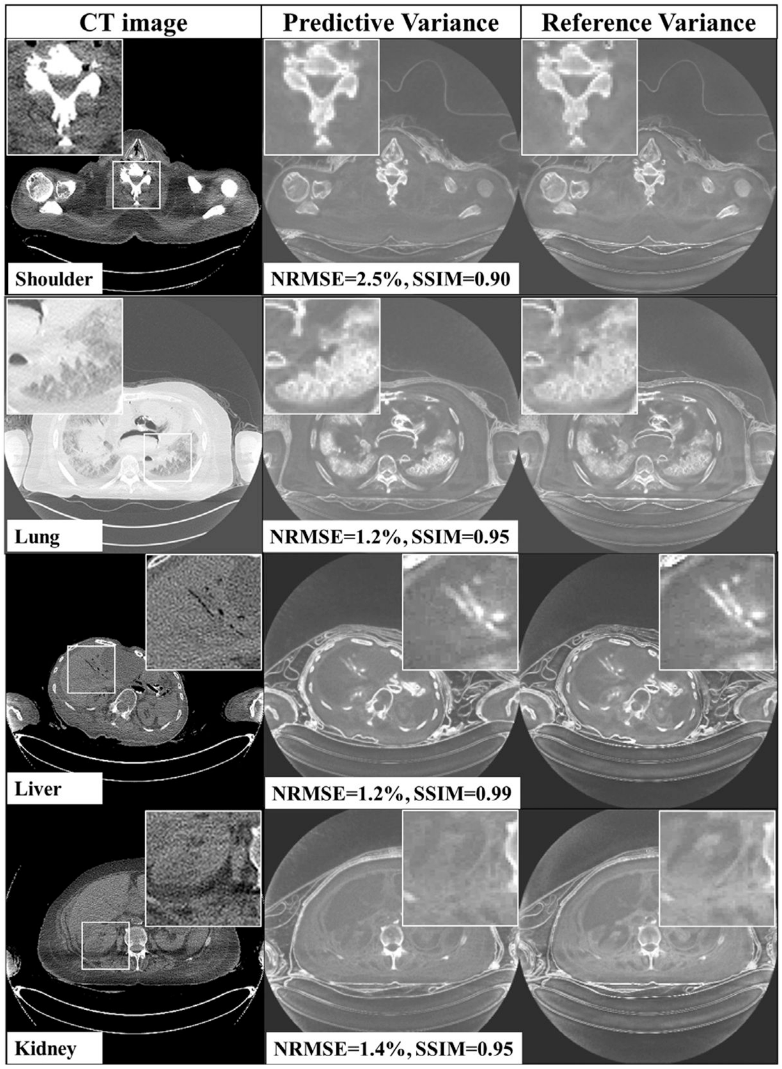

Fig. 2.

Predictive uncertainty agreed closely with the reference variance in testing cadaver images. The reference was measured from 100 repeated scans. The predictive variance was derived using one random scan and the proposed method, which generated 100 simulated realizations of inputs. No uncertainty recalibration was applied. NRMSE – normalized root-mean-square-error. SSIM – structural similarity index.

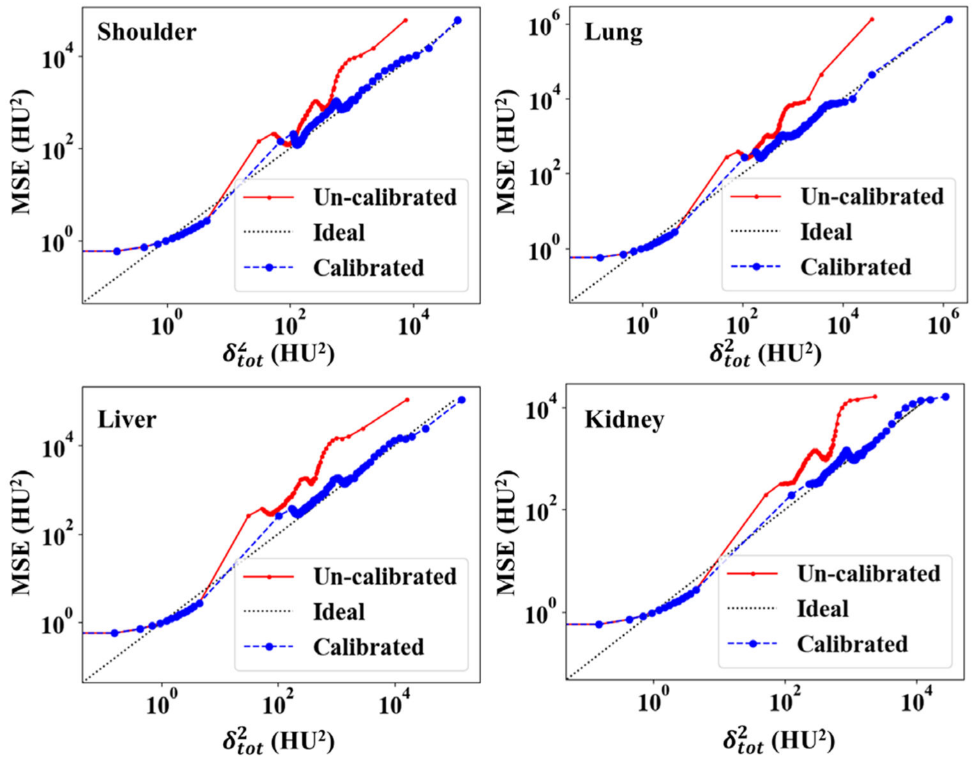

Fig. 3.

The proposed re-calibration method improved uncertainty calibration in testing images across four body regions. MSE – the expected error represented as mean-square-error. HU – Hounsfield unit.

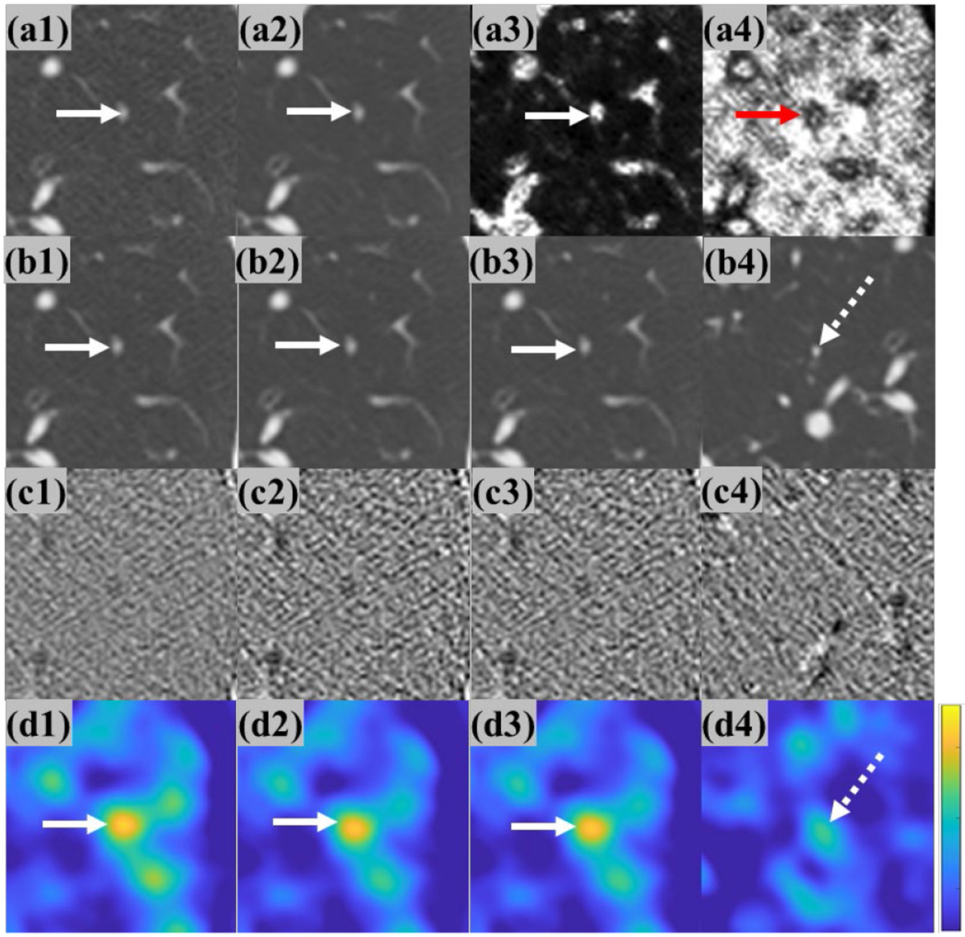

Fig. 4.

A patient case where DLMO correctly identified the true positive in ALF-derived images, while original pre-trained model led to a false positive. (a1) – (a4): Original noisy image, predictive mean, total uncertainty, and mean-uncertainty-ratio from the same testing patient case with a target part-solid nodule (3.4mm, 50% dose). Solid arrows indicate the target nodule locations. (b1) – (b3): Images from ALF with weighting functions (12) - (14), respectively, showing target nodules that were correctly localized by DLMO. (b4): The image from original pre-trained ResNet model for the same patient case, where the target nodule was missed and a false positive with lower DLMO response occurred. Dashed arrow indicates the false positive location. (c1) – (c4): difference images between (b1) – (b4) and the corresponding noisy inputs. (d1) – (d4): DLMO heatmaps for (b1) – (b4), respectively; brighter signals indicate higher likelihood of lesion presence. Display window (W/L): CT images 1600/-600 HU, difference images 100/0 HU. ALF – adaptive local fusion. DLMO – deep-learning model observer.

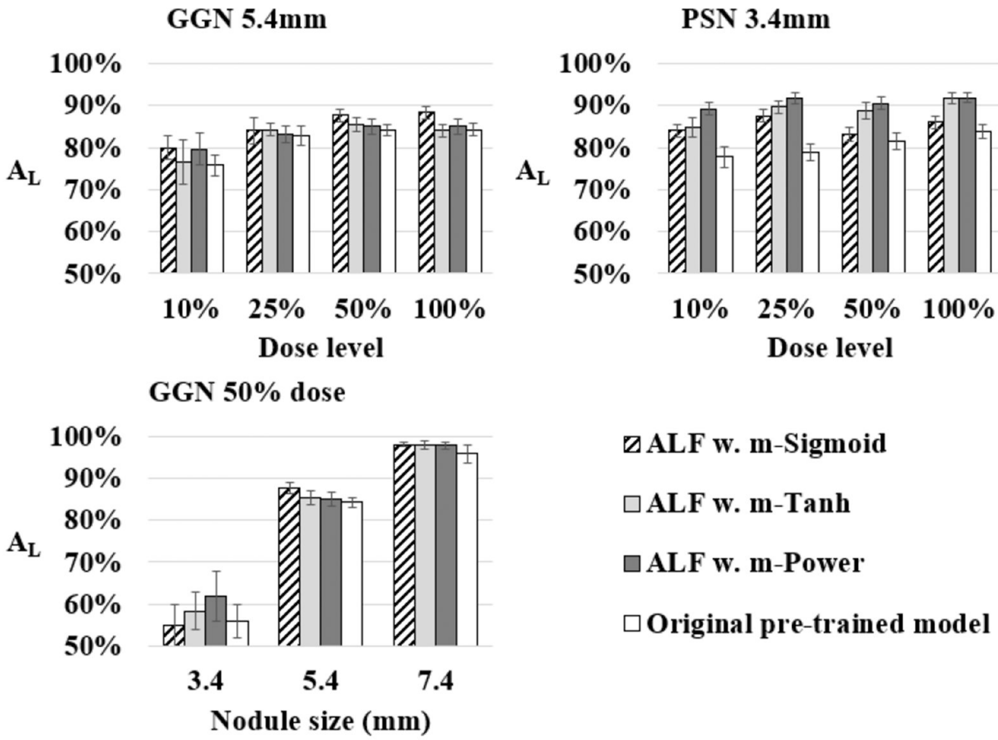

Fig. 5.

Nodule detectability in denoised images before and after applying the adaptive local fusion (ALF) in various conditions, involving 4 radiation dose levels (10% to 100%), 2 nodule types (GGN and PSN), 3 nodule size (diameter 3.4 to 7.4 mm). ALF used modified Sigmoid ("m-Sigmoid"), tanh ("m-Tanh"), and power function ("m-Power") in (12) - (14), separately. GGN – ground glass nodule. PSN – part-solid nodule. A_L – area under localization ROC.

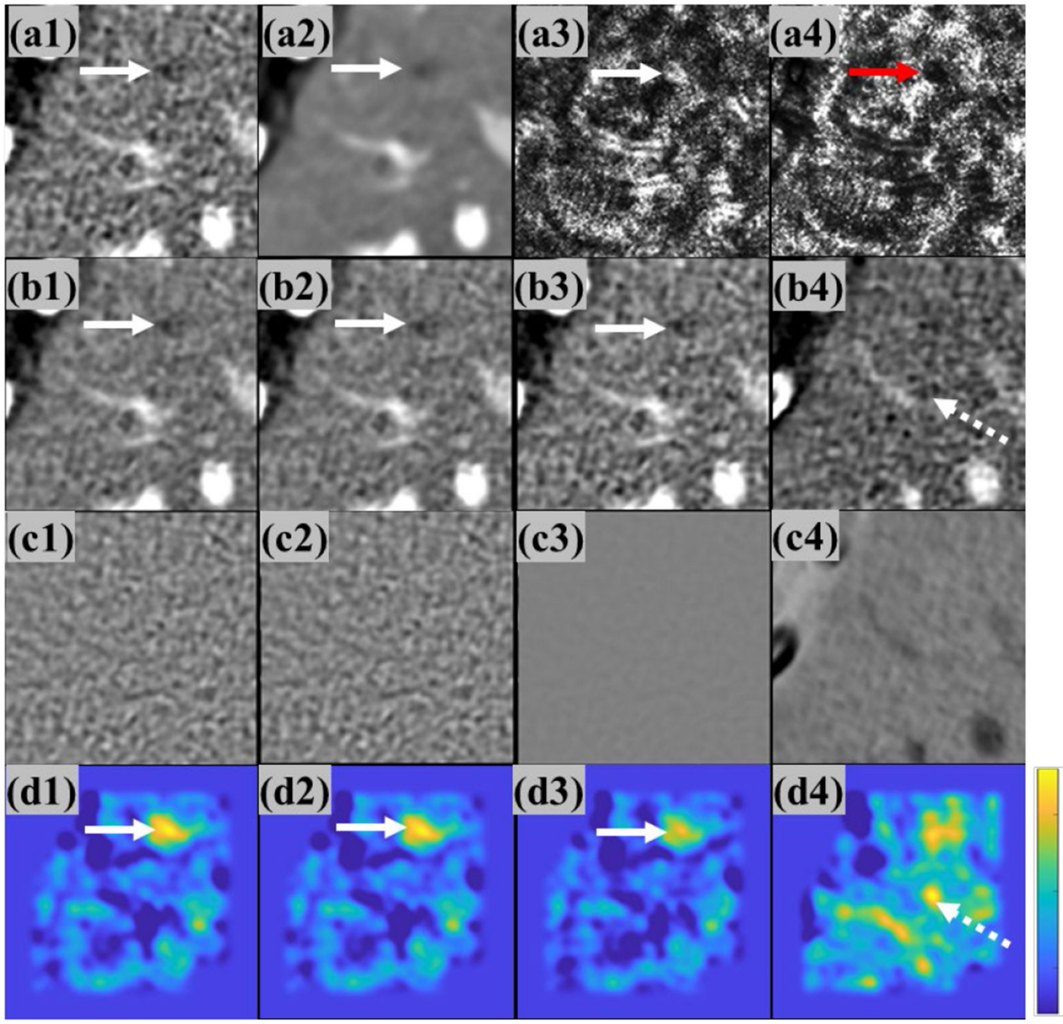

Fig. 6.

(a1) – (a4): Original noisy image, predictive mean, total uncertainty, and mean-uncertainty-ratio from the same testing patient case with a target low contrast liver lesion (7mm, 20HU, 50% dose). Solid arrows indicate target lesion locations. (b1) – (b3) Images from ALF with weighting functions (12) - (14), respectively, showing target lesions correctly localized by DLMO. (b4) The image from original pre-trained ResNet model for the same case, where target lesion was missed and a false positive with strong DLMO response occurred. Dashed arrow indicates the false positive location. (c1) – (c4) difference images between (b1) – (b4) and the corresponding noisy inputs. (d1) – (d4): DLMO heatmaps for (b1) – (b4), respectively. Brighter signals correspond to higher likelihood of lesion presence. Display window (W/L): CT images 212/120 HU, difference images 100/0 HU.

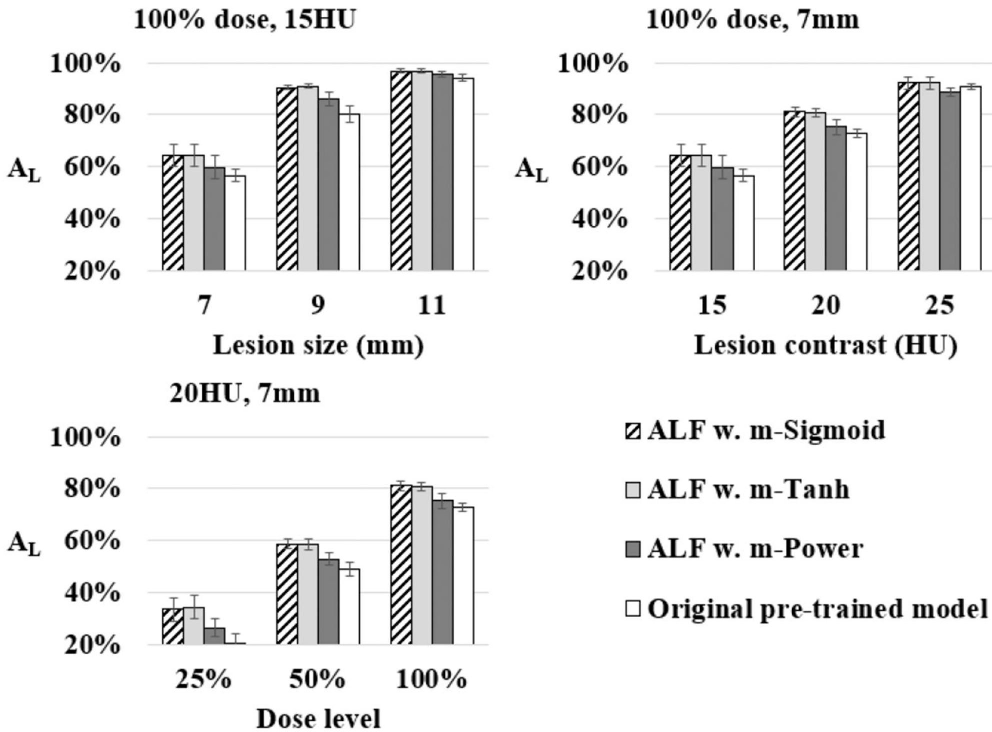

Fig. 7.

Liver lesion detectability in denoised images before and after applying the proposed adaptive local fusion (ALF) in various conditions, involving 3 radiation dose levels (25%, 50%, 100%), 3 size levels (diameter 7, 9, and 11 mm), 3 contrast levels (15, 20, and 25HU). ALF used modified Sigmoid (“m-Sigmoid”), tanh (“m-Tanh”), and power function (“m-Power”) in (12) - (14), separately. A_L – area under localization ROC.

TABLE I

QUANTITATIVE ASSESSMENT OF INFERENCE-TIME AUGMENTATION

	Shoulder	Lung	Liver	Kidney
NRMSE [a]	2.2%±0.2%	1.3%±0.6%	1.2%±0.1%	2.4%±0.1%
SSIM [b]	0.90±0.01	0.95±0.02	0.99±0.01	0.93±0.02

Quantitative metrics are represented as mean± standard deviation across all testing images per body region.

[a]Normalized root-mean-square-error

[b]Structural similarity index

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TABLE II

NORMALIZED CALIBRATION ERROR OF PREDICTIVE TOTAL UNCERTAINTY

	Shoulder	Lung	Liver	Kidney
Un-calibrated	5.5%	3.6%	6.5%	7.5%
Calibrated	1.3%	0.9%	1.7%	2.2%

Normalized calibration error depicted mean relative calibration error per bin. See Sec. II. C. for the details of the proposed calibration method.

Table III

CHEST CT IMAGE NOISE (HU) ACROSS DIFFERENT DOSE LEVELS

	10%	**25%**	**50%**	**100%**
Un-denoised	97.3±22.4	62.7±14.3	46.8±13.0	32.8±7.2
m-Sigmoid	48.6±6.4	36.3±6.5	29.2±9.0	21.3±4.7
m-Tanh	38.9±12.9	21.1±6.5	16.4±9.1	11.8±3.7
m-Power	29.4±3.6	23.5±3.0	21.3±7.8	16.7±3.5
No ALF	29.5±5.7	20.5±3.9	17.5±8.7	13.2±3.4

Image noise was measured in relatively uniform regions (e.g., muscle, fat)

“Un-denoised” denotes the original reconstruction (without deep-learning denoising) across the four dose levels

“m-Sigmoid”, “m-Tanh”, “m-Power” correspond to the ALF with weighting functions (12) - (14), respectively.

“No ALF” corresponds to the original pre-trained denoising model.

Table IV

ASSESSMENT OF SYSTEMATIC STRUCTURAL DISTORTION USING SSIM OF MEAN LUNG NODULE IMAGES ACROSS EXPERIMENTAL CONDITIONS

	5.4mm GGN 10%	5.4mm GGN 25%	5.4mm GGN 50%	5.4mm GGN 100%	3.4mm PSN 10%	3.4mm PSN 25%	3.4mm PSN 50%	3.4mm PSN 100%
Un-denoised	0.91	0.94	0.96	/	0.91	0.92	0.94	/
m-Sigmoid	0.94	0.96	0.96	0.96	0.92	0.92	0.94	0.94
m-Tanh	0.94	0.96	0.96	0.96	0.92	0.92	0.94	0.94
m-Power	0.94	0.96	0.96	0.96	0.92	0.92	0.94	0.94
No ALF	0.86	0.86	0.87	0.87	0.78	0.79	0.80	0.80

Structural similarity index (SSIM) was derived using the averaged lesion images across all lesion locations per experimental condition. The mean un-denoised lesion images at routine dose served as references. The 3.4mm and 7.4mm GGN were not involved in comparison due to lack of routine-dose references.

“Un-denoised” denotes the original reconstruction (without deep-learning denoising) across the four dose levels

“m-Sigmoid”, “m-Tanh”, “m-Power” correspond to the ALF with weighting functions (12) - (14), respectively.

“No ALF” corresponds to the original pre-trained denoising model.

Table V

LIVER CT IMAGE NOISE ACROSS DIFFERENT DOSE LEVELS

	25%	50%	100%
Un-denoised	28.3±8.6	20.0±5.6	14.0±3.8
m-Sigmoid	19.5±6.4	13.0±3.9	9.1±2.7
m-Tanh	19.4±6.1	13.0±3.8	9.3±2.7
m-Power	27.0±8.1	19.0±5.3	13.5±3.7
No ALF	26.0±8.1	18.0±5.2	12.6±3.5

Image noise was measured in relatively uniform liver parenchyma.

"Un-denoised" denotes the original reconstruction (without deep-learning denoising) across the four dose levels

"m-Sigmoid", "m-Tanh", "m-Power" correspond to the ALF with weighting functions (12) - (14), respectively.

"No ALF" corresponds to the original pre-trained denoising model.

Table VI

Assessment of systematic structural distortion using SSIM of mean liver lesion images across experimental conditions

	7mm 15HU 100%	7mm 20HU 100%	7mm 20HU 50%	7mm 20HU 25%	7mm 25HU 100%	9mm 15HU 100%	11mm 15HU 100%
Un-denoised	/	/	0.89	0.81	/	/	/
m-Sigmoid	0.99	0.99	0.92	0.86	0.99	0.98	0.98
m-Tanh	0.99	0.99	0.92	0.86	0.99	0.99	0.98
m-Power	0.99	0.99	0.89	0.82	0.99	0.99	0.98
No ALF	0.97	0.98	0.88	0.80	0.98	0.98	0.98

Structural similarity index (SSIM) was derived using the averaged lesion images across all lesion locations per experimental condition. Mean un-denoised routine-dose images served as references.

“Un-denoised” denotes the original reconstruction (without deep-learning denoising) across the four dose levels.

“m-Sigmoid”, “m-Tanh”, “m-Power” correspond to the ALF with weighting functions (12) - (14), respectively.

“No ALF” corresponds to the original pre-trained denoising model.